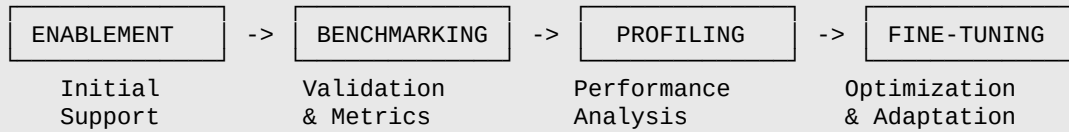


# Physical AI SDK - Model Lifecycle & Strategy 2026

*Hari Krishna Vydana*  
*PAVS AI*

# Model Lifecycle Overview



## Key Phases:

### 1. Enablement

- Get the model running
- Off-the-shelf installation (solve dependencies)
- benchmarking (respective metrics w.r.t.model)
- Identify the right dataset
- Fill the docs with the initial results and expectation

### 2. Benchmarking

- Measure baseline performance (task-metric, compute-metrics)
- Competitive analysis, With a right dataset on gpu vs cpu & gpu-npu
- work with CI/CD to automate it

### 3. Profiling

- Do compute analysis
- Find Hotspots and bottlenecks

### 4. Fine-tuning

- Explore opportunities to fix hotspots
- Implement the Model optimizations
- Benchmark the improvements in docs

# Progressive Model Support Strategy

## Phase 1: Individual Model Support

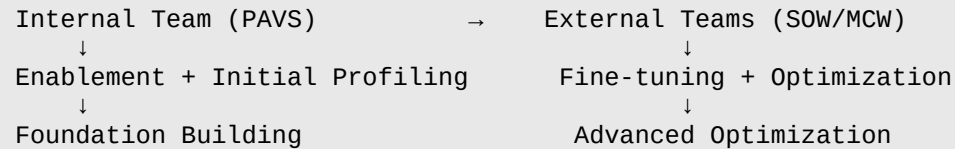
- Focus on single model enablement and optimization
- Establish baseline performance for each model
- Build reusable infrastructure and tooling
- Might get simple as we gain expertise

## Phase 2: Multi-Model Orchestration

- Systems: Orchestration of multiple models working together (More complex)
- System-level optimization and resource management will be critical
- End-to-end pipeline integration
- Will give a real feel of product

# Effective Team Collaboration

## Progressive Engagement Model



### Strategy:

**PAVS Team:** Handle enablement and initial profiling

- Establish working baseline
- Identify optimization opportunities
- Create clear specifications for external teams

**External Partners:** Focus on fine-tuning and advanced optimization

- Domain-specific optimization
- More quality time on Improvements
- Tangable and measurable outcomes

### Benefits:

- ✓ Better ROI on external resources
- ✓ Clear handoff criteria
- ✓ Faster time to production

# Model Status Across Lifecycle Stages

| Model Name      | Category               | Enablement | Profiling | Benchmarking | Fine-tuning |
|-----------------|------------------------|------------|-----------|--------------|-------------|
| MobileSAM       | Vision/Segmentation    | PAVS       | AIG       | AIG          | AIG         |
| YoloV12         | Object Detection       | PAVS       | PAVS      | PAVS         | MCW         |
| Qwen3-VL        | VLM                    | PAVS       | AIG       | AIG          | AIG         |
| LLaMA3.2-Vision | VLM                    | PAVS       | MCW       | MCW          | MCW         |
| Whisper         | ASR                    | PAVS       | MCW       | MCW          | MCW         |
| XTTS            | TTS                    | PAVS       | PAVS      | PAVS         | AIG         |
| DeepSeek-R1     | LLM                    | PAVS       | MCW       | MCW          | MCW         |
| Qwen2.5-VL      | VLM                    | PAVS       | AIG       | AIG          | AIG         |
| CrossFormer     | Vision Transformer     | PAVS       | MCW       | -            | -           |
| CenterPoint     | 3D Detection           | PAVS       | -         | -            | -           |
| OpenVLA         | Vision-Language-Action | PAVS       | -         | -            | -           |
| MedSigLIP       | Medical Classification | PAVS       | -         | -            | -           |
| EfficientNetV2  | Image Classification   | -          | -         | -            | -           |
| RAFT Stereo     | Stereo Matching        | -          | -         | -            | -           |

Legend: Complete | In-Progress | Not-Started

Key Insight: Start early, execute in parallel across multiple models,

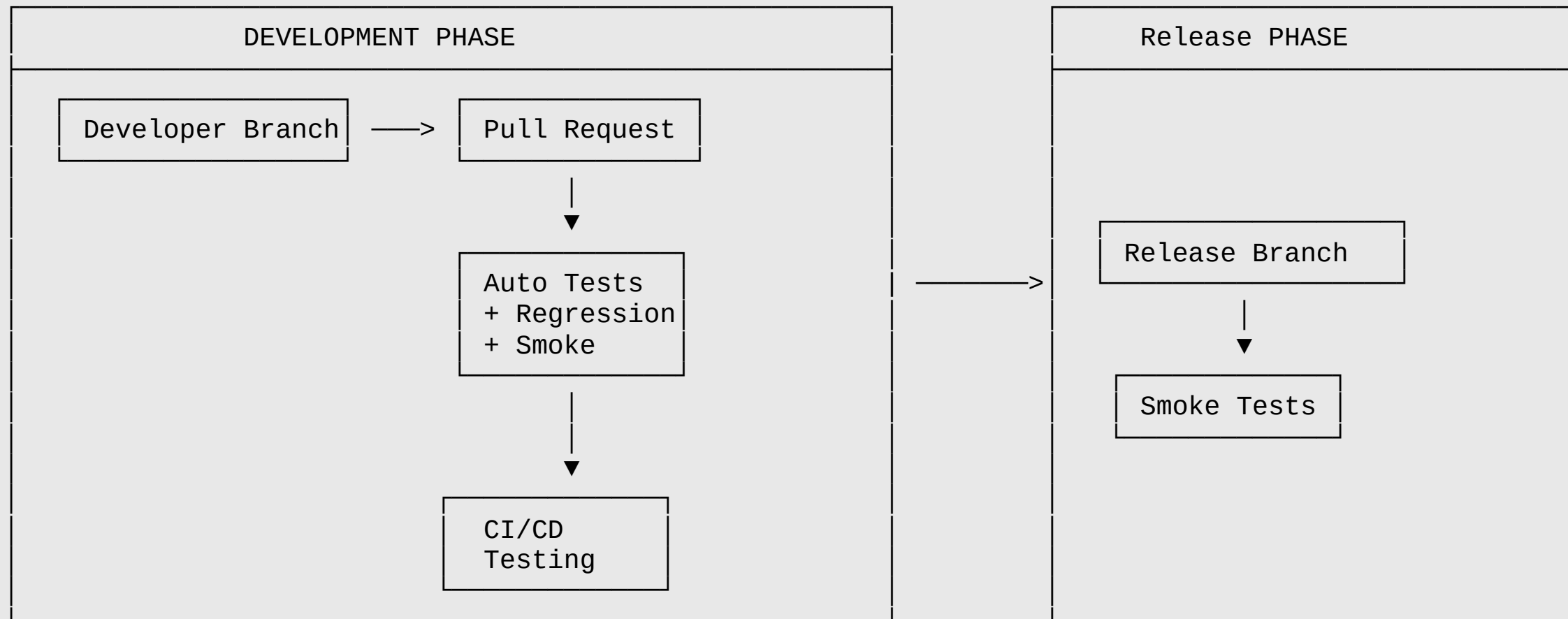
## Slide 6: Environment & CI/CD Infrastructure

### Environment Strategy:

```
physical-ai-sdk/  
├── models/  
│   ├── mobilesam/  
│   │   ├── gpu/  
│   │   │   ├── Dockerfile  
│   │   │   ├── pyproject.toml (uv)  
│   │   │   └── .venv/  
│   │   └── npu_gpu/  
│   │       ├── Dockerfile  
│   │       ├── pyproject.toml (uv)  
│   │       └── .venv/  
│   └── yolov12/  
│       ├── gpu/  
│       │   ├── Dockerfile  
│       │   ├── requirements.txt (pip)  
│       │   └── .venv/  
│       └── npu_gpu/  
│           ├── Dockerfile  
│           ├── requirements.txt (pip)  
│           └── .venv/  
└── llama3-vision/  
    ├── gpu/  
    │   ├── Dockerfile  
    │   ├── pyproject.toml (uv)  
    │   └── .venv/  
    └── npu_gpu/  
        ├── Dockerfile  
        ├── pyproject.toml (uv)  
        └── .venv/
```

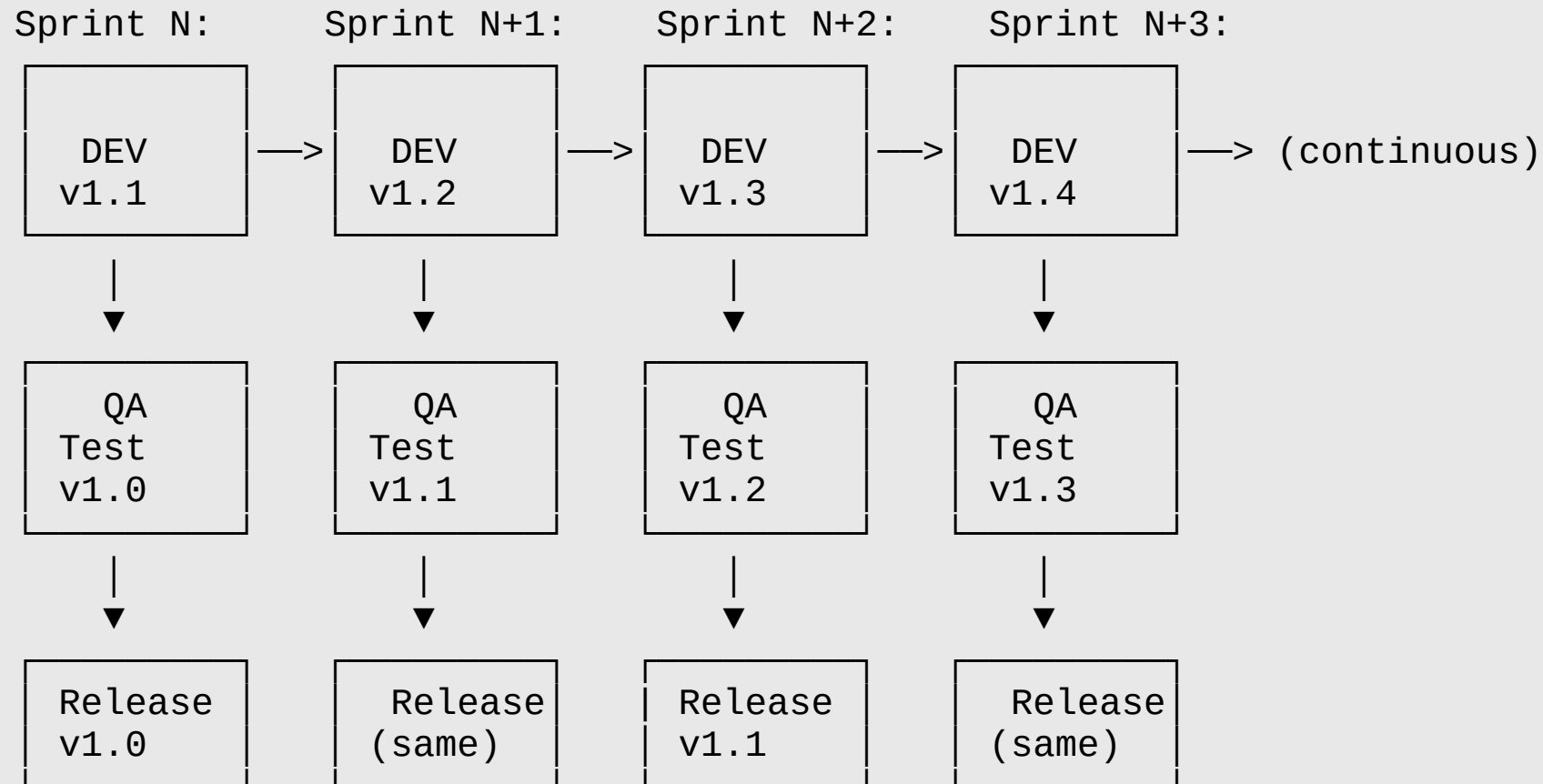
**Benefits:** ✓ Isolated dependencies | ✓ No version conflicts | ✓ Easy exploration

## CI/CD Pipeline:



## Dev - Release Timelines:

Sprint Timeline View:





## Slide 7: Container & Virtual Environment Options

### Option 1: Dockerfile

```
FROM rocm/pytorch:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
CMD ["python", "inference.py"]
```

### Option 2: uv + pyproject.toml

```
[project]
name = "mobilesam"
version = "1.0.0"
dependencies = [
    "torch>=2.0.0",
    "onnxruntime-rocm",
]
```

### Option 3: pip + requirements

```
torch==2.0.0
onnxruntime-rocm==1.17.0
opencv-python==4.8.0
```

## Model Distribution

- 📦 **\*\*HuggingFace Hub\*\*** - Pre-optimized ONNX/TensorRT, Quantized (INT8, FP16), ROCm builds
- 🔗 **\*\*Public Access\*\*** - Anyone can download and test
- ✅ **\*\*Verified Benchmarks\*\*** - Performance metrics included
- 📁 **\*\*Alternative:\*\*** GitHub Releases, Company registry, Cloud storage

# Expanding Model Portfolio

## Staying Current with Latest Open Source Models

### Emerging Neural Alternatives:

-  **End-to-End Neural Pipelines** replacing traditional CV pipelines
  - Example: Neural speech-to-speech model
  - Example: VLA++, pi-0++
-  **Latest Model Categories to Track:**
  - Multimodal Foundation Models (GPT-4V alternatives)
  - Action-VLMs (RT-2, OpenVLA successors)
  - Neural Rendering (NeRF, Gaussian Splatting for robotics)
  - On-device Edge LLMs (< 3B parameters)

### Strategy:

1. Monthly scan of arxiv.org, HuggingFace, GitHub trending
2. Quick feasibility assessment (2-3 days cycle)
3. Add to roadmap if aligned with Physical AI use cases
4. If the model fits in requirements and aligns with our business directions (Robotics, healthcare, Industrial we can add it)

# Benchmarking & Documentation Strategy

## Documentation as Code:

```
docs/
├── index.md
├── models/
│   ├── mobilesam.md
│   ├── yolov12.md
│   └── benchmarks/
├── guides/
│   ├── quick-start.md
│   └── deployment.md
└── _config.yml
```

## Automated Workflow:

1. **Code + Benchmarks** → GitHub Repo
2. **CI/CD** → Auto-generate tests, run benchmarks
3. **GitHub Pages** → Publish automatically

## Benefits:

- ✓ Version-controlled documentation
- ✓ Automated benchmark updates
- ✓ Professional presentation
- ✓ Community collaboration

# Pages Example - Mock Preview

## Sample Documentation Site Layout

### Homepage Preview:

Physical AI SDK Documentation

🚀 Quick Start

📊 Benchmarks

🤖 Models

📖 Model Catalog

Vision Models (12)

MobileSAM, YoloV12, EfficientNet...

VLM Models (5)

Qwen3-VL, LLaMA3.2-Vision...

Audio Models (3)

Whisper, XTTS...

📈 Performance Dashboard

| Model     | FPS      | Latency | Acc |
|-----------|----------|---------|-----|
| MobileSAM | 240fps   | 4.2ms   | 94% |
| YoloV12   | 240fps   | 4.2ms   | 92% |
| Whisper   | 100tok/s | 10ms    | 96% |

# Practical Example - Model Lifecycle

## Case Study: Qwen3-VL Integration

### Enablement:

-  Model weights download and conversion
-  Basic ONNX-RT integration
-  Simple inference test (single image)
-  Docker environment setup

### Profiling:

-  Layer-wise performance analysis
-  Memory usage tracking
-  Hotspot identification (Conv3D, GEMM)

### Benchmarking:

-  Performance on dataset vs leaderboard
-  KPI testing: Target 30fps @ 1080p
-  Accuracy validation on VQA datasets
-  Power consumption measurement
-  Automate the benchmarking setup

### Fine-tuning:

-  Kernel optimization for GEMM
-  INT8 quantization (W8A8)
-  Automotive dataset fine-tuning

# Resource Management Strategy

## Balancing Internal & External Resources

### Resource Allocation Matrix:

| Lifecycle Stage | Internal Team (PAVS)                                                                        | External Partners                                                                            | Max.Exptd Duration |
|-----------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------|
| Enablement      |  (Primary) |  (Support) | 1 weeks            |
| Profiling       |  (Lead)    |  (Assist)  | 3 days             |
| Benchmarking    |  (Guide)   |  (Execute) | 3 days             |
| Fine-tuning     |  (Review)  |  (Primary) | 4-8 weeks          |

### Handoff Criteria:

- ✓ Model runs successfully on target hardware
- ✓ Profiling report completed
- ✓ Baseline benchmarks established
- ✓ Optimization opportunities documented, explored

# Success Metrics & KPIs

## Measuring Lifecycle Effectiveness

### Model Lifecycle KPIs:

| Metric                        | Target         | Current | Trend        |
|-------------------------------|----------------|---------|--------------|
| Average Time to Enablement    | < 2 weeks      | 1 weeks | ⬇️ Improving |
| Models in Profiling Stage     | 5-8 concurrent | 6       | ➡️ Stable    |
| Benchmark Pass Rate (1st try) | > 80%          | 75%     | ⬆️ Improving |
| Fine-tuning Speedup (avg)     | > 2x           | 2.3x    | ⬆️ Good      |
| CI/CD Pipeline Success Rate   | > 95%          | 97%     | ✅ Excellent  |

### Portfolio Metrics:

- Total Models Supported: **45+**
- Production-Ready Models: **28**
- Models in Development: **12**
- Experimental/Research: **5**

## Roadmap & Next Steps

### Q1 2026 (Current):

-  Establish lifecycle framework
-  Complete \* models through full lifecycle
-  Deploy GitHub Pages documentation

### Q2 2026:

-  Expand to 15 release-ready models
-  Implement automated benchmarking CI/CD
-  Start 1 orchestration systems

### Q3 2026:

-  Multi-model orchestration (2 pipelines)
-  Add 10 emerging models from watchlist
-  Scale fine-tuning operations

### Q4 2026:

-  Production deployment for 3 segments
-  2027 strategy planning